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ATAS - Academic Text Analysis System

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Software

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Abstract

The **ATAS – Academic Text Analysis System** (Brazilian Portuguese **SATA – Sistema de Análise de Textos Acadêmicos**) is open-source software to support **textual content analysis**. Inspired by **Bardin's method** (Bardin, 2020), ATAS streamlines **extraction, filtering, and statistical analysis** of academic texts to surface semantic and linguistic patterns. It is particularly useful to **Humanities and Social Sciences (Geography)** researchers, offering tools for **keywords, bigrams, lexical categories**, and other quantitative text measures. ATAS integrates **NLP** workflows and exports data to tools such as **Gephi** for semantic network analysis (Bastian et al., 2009).

The toolset is localized for **Brazilian Portuguese** and is actively used by the **GETE – Grupo de Estudos Territoriais (Territorial Studies Group)** at **UEPG** and the **GEPES – Grupo de Pesquisa Redes de Poder, Migrações e Dinâmicas Territoriais (Research Group on Power Networks, Migrations, and Territorial Dynamics)** at **UNICENTRO** (Brazil), both operating in **Master's and PhD** programs.

Statement of Need

In Brazil's Humanities and Social Sciences (Geography), many scholars still rely on manual or semi-manual workflows to process large textual corpora. Historical barriers—limited programming experience, scarce infrastructure, and uneven training—have hindered the adoption of computational methods (Metzler et al., 2016). At the same time, **Large Language Models (LLMs)** have transformed NLP in recent years, opening powerful possibilities but also raising concerns about **language coverage, transparency, and reproducibility** (OpenAI, 2023; Scao et al., 2022). For **Portuguese (Brazil)** specifically, the ecosystem has improved with **encoder models** such as **BERTimbau** (Souza et al., 2020) and more recent **PT-BR Encoders** (Mello et al., 2024), yet accessible, GUI-driven tools that operationalize these advances for **non-specialists** remain scarce.

ATAS addresses this gap as an **open-source, GUI-based** system that automates key tasks—keyword filtering, bigram networks, lexical metrics—**without requiring advanced coding**. Localization to **Brazilian Portuguese** promotes equitable access to computational analysis and supports reproducible research pipelines for Human Geography and allied fields.

Beyond facilitating content analysis, ATAS helps investigate how **discourse constitutes space**. In Human Geography, **space** is socially produced and relational rather than a neutral container (Lefebvre, 1991; Massey, 2005). While GIS excels at mapping and spatial querying, its discrete data structures can under-represent **processual and discursive spatialities**. ATAS complements cartography by extracting **semantic networks and entities** directly from text, enabling researchers to track how places, actors, and relations are constructed, contested, and reconfigured in academic and policy discourse.

Features and Usage

1. Text Filtering (Brazilian Portuguese: Filtragem de Texto)

Extracts verbs, adjectives, and nouns from texts, facilitating qualitative analyses.

- **Library used:** `spaCy` (Explosion AI, 2023)
- **How to use:**
  1. Open ATAS and go to the *Filter Text* option.
  2. Select the `.txt` file to be analyzed.
  3. The system processes the text and saves a new filtered file.

2. Table Conversion (Brazilian Portuguese: Conversão para Tabela)

Generates bigrams from the text and exports the data in CSV format, useful for analysis in Gephi.

- **Library used:** `pandas` (The Pandas Development Team, 2023)
- **How to use:**
  1. Access the *Convert Text to Table* option.
  2. Choose a `.txt` file.
  3. ATAS generates a CSV file containing the bigrams, ready for network analysis.

3. Gender Identification (Brazilian Portuguese: Identificação de Gênero)

Automatically classifies the gender of proper names found in a textual dataset.

- **Library used:** `gender_guesser` (Hutchinson, 2016)
- **How to use:**
  1. Select the *Identify Gender* option.
  2. Upload a CSV file containing a list of names.
  3. ATAS generates a new CSV with the gender classification associated with each name.

4. Text Statistics (Brazilian Portuguese: Estatísticas de Texto)

Provides quantitative metrics such as **word frequency, named entities, and lexical diversity** to surface thematic emphases, recurring actors, and stylistic features in academic texts.

- **Libraries used:** `spaCy`, `pandas`
- **How to use:**
  1. Go to the *Text Statistics* option.
  2. Select a text file.
  3. The system presents a detailed statistical report, including **word clouds and graphs**.

Note: Sentiment analysis will be integrated in future releases with **Portuguese-specific models** (e.g., Stanza, NLPNet, UDPipe) and, where appropriate, **open LLMs** fine-tuned for PT-BR (Touvron et al., 2023), while preserving ATAS's principles of **openness, transparency, and independence from proprietary APIs** (OpenAI, 2023; Scao et al., 2022).

5. Graphical Interface

ATAS offers an intuitive visual interface based on `tkinter` and `ttkbootstrap`, allowing users without programming knowledge to easily access its functionalities.

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Installation

ATAS can be installed locally from source. The repository includes all dependencies in the `requirements.txt` file.

```
git clone https://github.com/AlidesChimin/SATA.git
cd SATA
pip install -r requirements.txt
python main.py
```

The software was tested under **Python 3.8+** on **Linux and Windows** environments.

Reproducibility and Testing

All examples in the paper can be reproduced using the sample datasets available in the `/examples` directory.
Each function (e.g., text filtering, bigram extraction, gender identification) can be validated independently.
Automated tests are implemented through continuous integration on **GitHub Actions**, confirming successful imports, dependencies, and environment reproducibility.

Community Guidelines

The project follows open-source contribution standards.
Users and contributors can: 1. Open issues and pull requests on GitHub. 2. Follow the guidelines described in the `CONTRIBUTING.md` file. 3. Adhere to the `CODE_OF_CONDUCT.md` included in the repository.

Scholarly Effort

ATAS comprises approximately **910 total lines of source code**, of which **655 are Python logic**, distributed across eight modules.
The system includes **38 functions** and **6 classes**, integrating major open-source libraries (`spaCy`, `pandas`, `tkinter`, `ttkbootstrap`, `gender_guesser`, and `PIL`).
It was developed by **Dr. Alides Baptista Chimin Junior** as part of ongoing research in the **GEPES** and **GETE** groups, within the field of **Human Geography**.
The project reflects over a year of design, implementation, and integration efforts, bridging computational linguistics and content analysis for the Humanities.

Implementation

ATAS is developed in **Python 3.8+** and utilizes:

- **`spaCy`** for text processing.
- **`pandas`** for data manipulation.
- **`tkinter`** and **`ttkbootstrap`** for the graphical interface.
- **`gender_guesser`** for gender identification.

While the current version relies primarily on `spaCy`, future developments will add **Stanza/NLP-Net/UDPipe** backends and optional **open LLM** components for PT-BR tasks, keeping the stack reproducible and OSI-compliant. The source code is available on GitHub: https://github.com/AlidesChimin/SATA.

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